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Research article

Pigeon interaction mode switch-based UAV distributed flocking control under obstacle environments[☆]Huaxin Qiu, Haibin Duan^{*}

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ABSTRACT

Unmanned aerial vehicle (UAV) flocking control is a serious and challenging problem due to local interactions and changing environments. In this paper, a pigeon flocking model and a pigeon coordinated obstacle-avoiding model are proposed based on a behavior that pigeon flocks will switch between hierarchical and egalitarian interaction mode at different flight phases. Owing to the similarity between bird flocks and UAV swarms in essence, a distributed flocking control algorithm based on the proposed pigeon flocking and coordinated obstacle-avoiding models is designed to coordinate a heterogeneous UAV swarm to fly through obstacle environments with few informed individuals. The comparative simulation results are elaborated to show the feasibility, validity and superiority of our proposed algorithm.

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1. Introduction

The unmanned aerial vehicle (UAV), an aircraft without a human pilot aboard, has received extensive attentions from researchers owing to superior performance in missions that are too "dull, dirty or dangerous" for humans [1–3]. According to Lanchester's laws, the number of UAVs is a more important factor to affect the combat power of a military force than the capability. Therefore, the usage of UAV swarms will be a better choice to control battle-field situations. Getting groups of UAVs to a swarm will require the design of distributed flocking control algorithms without huge computational overheads or large communication bandwidths.

Except traditional control algorithms [4–6], the swarm intelligence in multifarious complex systems, especially in bird flocks, has provided a novel but feasible thought for the UAV distributed flocking control due to their similarities: the stigmergy, no-center, simple individual and self-organization [7,8]. In the literature, a few researchers have made some exploratory attempts. Hauert et al. [9] demonstrated Reynolds' Boid model, the first flocking model which reveals the basic rules of bird flocking,

with a swarm of small fixed-wing UAVs in outdoor experiments. Vásárhelyi et al. [10] implemented fundamental collective flight tasks with a multi-copter swarm based on the control algorithm which has a structure analogous to that of the Boid model. Saska [11] introduced a novel stabilization approach via the Boid model for a UAV swarm along a predefined path through obstacle environments. However, these methods have not considered the possibility that a swarm might consist of individuals with different capability levels.

As an added bonus, the problem has already appeared on the table of some researchers who are concerned about the collective motion mechanism of bird flocks. Ballerini et al. [12] discovered that each bird interacts on average with a fixed number of neighbors based on the egalitarian mode. Nagy et al. [13,14] found a hierarchical leadership network from multiple pigeon flock flights. In the hierarchy, pigeons follow the individuals in upper ranks and lead the flights of individuals in lower ranks. The stable leadership hierarchy is inferred to be the resultant of group coordination based on different individual competences [15]. Zhang et al. [16] discovered that pigeon flocks adopt a switching strategy between the hierarchy and egalitarian interaction modes. Each pigeon tends to follow the average of neighbors during smooth flights, whereas it switches to follow its leaders when sudden turns or zigzags occur.

This paper focuses on the distributed flocking control algorithm design for a heterogeneous UAV swarm under obstacle environments based on the pigeon interaction mode switch behavior. Firstly a pigeon flocking model is proposed based on the artificial

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potential field method to allow a pigeon flock to safely fly to a target under obstacle environments. When flying based on the proposed flocking model, individuals may fall into local minimums of the potential field before reaching the final target. Fortunately, individuals are available to escape the local minimum by setting appropriate temporary targets [17]. Considering the different competences to avoid obstacles, a pigeon coordinated obstacle-avoiding model is presented based on the pigeon interaction mode switch behavior. Individuals equipped with more information about temporary targets will locate at more upper ranks in the hierarchy. Owing to the similarity between pigeon flocks and UAV swarms in mission requirements, the pigeon flocking and coordinated obstacle-avoiding models are applied to the control field to coordinate a heterogeneous UAV swarm to fly through a complex environment with obstacles. Some dynamic constraints of UAVs are considered to improve the practicality of the UAV distributed flocking control algorithm.

The rest of the paper is organized as follows. Section 2 models the collective motion of a pigeon flock to safely fly to a target under obstacle environments. Section 3 presents a coordinated obstacle-avoiding model based on the pigeon interaction mode switch behavior in consideration of the different individual obstacle-avoiding competence. Section 4 describes a UAV model. Section 5 proposes a distributed flocking control algorithm for a UAV swarm based on the pigeon flocking and coordinated obstacle-avoiding models. Comparative simulations are elaborated in Section 6, and our concluding remarks are drawn in Section 7.

2. Pigeon flocking model

The organized flight of pigeon flocks in the sky is seemingly a well-rehearsed group dance. The complex collective behavior arises from simple local interaction rules that an individual conducts cohesion, alignment and separation based on the states of its flockmates. The flockmate selection are considered to involve two interaction modes: the egalitarian interaction mode and hierarchical interaction mode. In the former mode, each pigeon interacts within a fixed neighborhood region (FNR) or with a fixed number of neighbors (FNN). In the latter mode, each pigeon follows the individuals with upper ranks in the hierarchical leadership network. A switching strategy is employed by each pigeon to balance the two apparently contradictory interaction modes: Each pigeon adopts the egalitarian interaction mode during smooth flights, whereas it switches to the hierarchical interaction mode when sudden turns or zigzags occur. In this section, a pigeon model is proposed without regard for the switching strategy in pigeon flocks.

Consider N pigeons flying in Euclidean 3-spaces with M obstacles, and the dynamic model of each individual regarded as one mass point is as the following equation:

$$\begin{cases} \dot{\mathbf{X}}^i = \mathbf{v}^i \\ m^i \dot{\mathbf{v}}^i = \mathbf{u}^i - k^i \mathbf{v}^i, i = 1, \dots, N \end{cases} \quad (1)$$

where $m^i > 0$ is the mass of each individual and set to be 1kg in this paper, $\mathbf{X}^i, \mathbf{v}^i, \mathbf{u}^i \in \mathbb{R}^3$ are the position vector, velocity vector and control input of each individual respectively, $k^i > 0$ is the gain of velocity attenuation and $-k^i \mathbf{v}^i$ is the velocity damping which is equivalent to air resistances.

The control input of each individual is as the following equation:

$$\mathbf{u}_l^i = \begin{cases} f_f + f_t + f_o + f_c + f_{a_vn} + f_{a_ve} + k^i \mathbf{v}_l^i, & l = 1, 2 \\ f_{a_he} + f_{a_ve} + k^i \mathbf{v}_l^i, & l = 3 \end{cases} \quad (2)$$

where $l = 1, 2, 3$ respectively correspond to the x -axis, y -axis and z -axis directions of the inertial coordinate system and the formation control component f_f is as the following equation:

$$f_f = K_f \left(\frac{\sum_{j \in \{\|\mathbf{X}_{1,2}^{ji}\| \leq R_{comm}\}} w_j \nabla \|\mathbf{X}_{1,2}^{ji}\| P_f^{ji}}{\sum_{j \in \{\|\mathbf{X}_{1,2}^{ji}\| \leq R_{comm}\}} w_j} \right) \quad (3)$$

where $K_f > 0$ is the gain of formation control, $\|\mathbf{X}_{1,2}^{ji}\|$ is the horizontal distance between individual i and j , w_j is the influence weight of individual j to individual i , R_{comm} is the horizontal communication range and the potential function P_f^{ji} for formation control in this paper is as the following equation:

$$P_f^{ji}(\|\mathbf{X}_{1,2}^{ji}\|) = \frac{1}{2} \|\mathbf{X}_{1,2}^{ji}\|^2 - (R_{desire})^2 \cdot \ln \|\mathbf{X}_{1,2}^{ji}\| \quad (4)$$

where R_{desire} is the expected horizontal distance between individual i and j .

The migration control component f_t is as the following equation which leads each individual to target T :

$$f_t = \begin{cases} K_t \nabla_{(\mathbf{T}_l - \mathbf{X}_l^i)} P_t, & \|\mathbf{T}_l - \mathbf{X}_l^i\| > R_{lim1} \\ 0, & \|\mathbf{T}_l - \mathbf{X}_l^i\| \leq R_{lim1} \end{cases} \quad (5)$$

where $K_t > 0$ is the attraction gain of the target, the potential function P_t of the target in this paper is $\|\mathbf{T}_l - \mathbf{X}_l^i\|^3/3$ and R_{lim1} is the limited distance of potential field influence of the target.

The control component f_o of obstacle avoidance is as the following equation:

$$f_o = K_o \sum_{j' \in \{\rho(\mathbf{X}^i, \mathbf{obs}_{j'}) \leq R_{sense}\}} \nabla_{\mathbf{X}^i} P_o^{j'} \quad (6)$$

where $K_o > 0$ is the repulsion gain of obstacles, $\rho(\mathbf{X}^i, \mathbf{obs}_{j'})$ is the shortest distance between individual i and obstacle j' , $\mathbf{obs}_{j'}$ is the position vector of the point closest to individual i on the surface of obstacle j' , R_{sense} is the limited distance of potential field influence of the obstacle j' and the potential function $P_o^{j'}$ of the obstacle j' in this paper is as the following equation:

$$P_o^{j'} = -\frac{1}{\rho(\mathbf{X}^i, \mathbf{obs}_{j'})} - \frac{2}{R_{sense}} \ln(\rho(\mathbf{X}^i, \mathbf{obs}_{j'})) + \frac{1}{(R_{sense})^2} \rho(\mathbf{X}^i, \mathbf{obs}_{j'}) \quad (7)$$

The following equation can be obtained by substituting Eq. (7) into Eq. (6):

$$f_o = K_o \sum_{j' \in \{\rho(\mathbf{X}^i, \mathbf{obs}_{j'}) \leq R_{sense}\}} \left(\frac{1}{\rho(\mathbf{X}^i, \mathbf{obs}_{j'})} - \frac{1}{R_{sense}} \right) \frac{\partial \rho(\mathbf{X}^i, \mathbf{obs}_{j'})}{\partial \mathbf{X}^i} \quad (8)$$

where $\frac{\partial \rho(\mathbf{X}^i, \mathbf{obs}_{j'})}{\partial \mathbf{X}^i}$ can be represented as

$$\frac{\partial \rho(\mathbf{X}^i, \mathbf{obs}_{j'})}{\partial \mathbf{X}^i} = \frac{\mathbf{X}^i - \mathbf{obs}_{j'}}{\rho(\mathbf{X}^i, \mathbf{obs}_{j'})} \quad (9)$$

The control component f_c is to avoid the collision between individuals. In this paper, the expression of f_c is similar to f_o :

$$f_c = K_c \sum_{j \in \{\|X_{i,2}^{ij}\| \leq R_{lim2}\}} \left(\frac{1}{\|X_{i,2}^{ij}\|} - \frac{1}{R_{lim2}} \right)^2 \frac{\partial \|X_{i,2}^{ij}\|}{\partial X^i} \quad (10)$$

where $K_c > 0$ is the repulsion gain of individual j , $\frac{\partial \|X_{i,2}^{ij}\|}{\partial X^i}$ can be represented as $\frac{X^i - X^j}{\|X_{i,2}^{ij}\|}$ and R_{lim2} is the limit distance of repulsion potential field influence of the individual j .

The alignment control component f_{a_vn} for the velocities of neighbors is as the following equation:

$$f_{a_vn} = K_{a_vn} \left[\frac{\sum_{j \in \{\|X_{i,2}^{ij}\| \leq R_{comm}\}} w_j (\mathbf{v}_j - \mathbf{v}_i)}{\sum_{j \in \{\|X_{i,2}^{ij}\| \leq R_{comm}\}} w_j} \right] \quad (11)$$

where $K_{a_vn} > 0$ is the alignment gain of neighbors' velocities.

The alignment control component $f_{a_ve}^l$ between desired velocity $\mathbf{v}_e$ and individual velocity $\mathbf{v}_i$ in the l^{th} dimension of the Euclidean space is as the following equation:

$$f_{a_ve}^l = K_{a_ve}^l (\mathbf{v}_e - \mathbf{v}_i) \quad (12)$$

where $K_{a_ve}^l > 0$ is the alignment gain of the desired velocity.

The alignment control component f_{a_he} between the target altitude T_3 and individual altitude X_3 is as the following equation:

$$f_{a_he} = K_{a_he} (T_3 - X_3) \quad (13)$$

where $K_{a_he} > 0$ is the alignment gain of the target altitude.

3. Pigeon coordinated obstacle-avoiding model

When flying in a pigeon flock through a complex environment with obstacles based on the above flocking model, individuals may fall into a local minimum before reaching final target $\mathbf{Tf}$. If individuals are informed about how to set appropriate temporary targets, they will successfully escape the local minimum. In this paper, a collective obstacle-avoiding model is proposed based on the pigeon interaction mode switch behavior to coordinate a

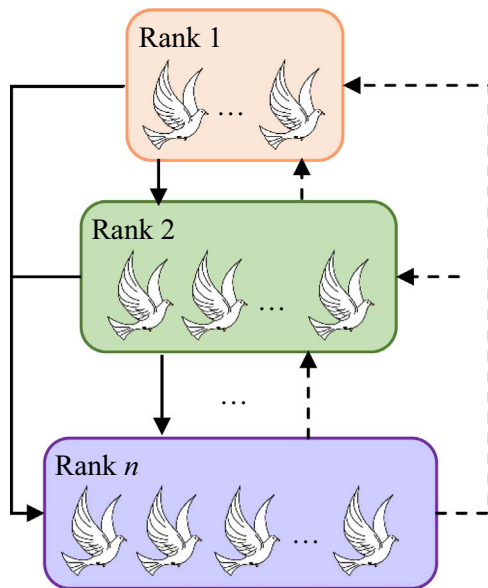


Fig. 1. Hierarchical structures in pigeon flocks.

pigeon flock to flight though the obstacle environment with few informed individuals. Specific steps of the coordinated obstacle-avoiding model are as follows:

Initialization: Identifiers $Flag_{obs}^i$, $Flag_{trap}^i$ and $Flag_{uninfor}^i$ are set to be 0. As shown in Fig. 1, individuals in the pigeon flock are divided into different ranks $Rank$ in a hierarchical structure. The individuals in the upper rank are equipped more information about how to set a temporary target to avoid getting into a local minimum. As shown in Fig. 2, if the long side L_j' of obstacle j' is parallel to y-axis and $\mathbf{Tf}_1 > \mathbf{X}_1^i$, the location of the temporary target $\mathbf{Tt}^j$ is as the following equation:

$$\mathbf{Tt}^j = \begin{cases} (C_1^j - W_j/2, & j' \in \text{Case 1 and } X^i \in \text{Area 1} \\ C_2^j + L_j'/2 + \delta_j', \mathbf{Tf}_3), \\ (C_1^j - W_j/2, & j' \in \text{Case 1 and } X^i \in \text{Area 2} \\ C_2^j - L_j'/2 - \delta_j', \mathbf{Tf}_3), \\ (C_1^j, C_2^j + L_j' + \delta_j', \mathbf{Tf}_3), & j' \in \text{Case 2 and } \alpha_j = 3\pi/2 \\ (C_1^j, C_2^j - L_j' - \delta_j', \mathbf{Tf}_3), & j' \in \text{Case 2 and } \alpha_j = \pi/2 \\ (C_1^j - L_j' - \delta_j', & j' \in \text{Case 3 and } X^i \in \text{Area 1} \\ C_2^j + L_j'/2 + R_{sense}, \mathbf{Tf}_3), \\ (C_1^j, C_2^j + L_j'/2 + \delta_j', \mathbf{Tf}_3), & j' \in \text{Case 3 and } X_2^i \geq C_2^j + L_j'/2 + R_{sense} \\ (C_1^j - L_j' - \delta_j', & j' \in \text{Case 3 and } X^i \in \text{Area 2} \\ C_2^j - L_j'/2 - R_{sense}, \mathbf{Tf}_3), \\ (C_1^j, C_2^j - L_j'/2 - \delta_j', \mathbf{Tf}_3), & j' \in \text{Case 3 and } X_2^i \leq C_2^j - L_j'/2 - R_{sense} \end{cases} \quad (14)$$

where C^j is the center of the obstacle j' in the xy plane, L_j' and L_j'' are the length of the obstacle j' in the y-axis and x-axis directions respectively, W_j is the width of the obstacle j' , α_j is the positive angle from L_j'' to L_j' , and δ_j is a setting parameter of temporary targets and greater than R_{sense} .

Step 1: If $Flag_{obs}^i \neq 0$, $Flag_{trap}^i = 1$. If individual i have passed the artificial potential influence area of the obstacle j' , $Flag_{trap}^i = 0$.

Step 2: If $Flag_{trap}^i = 1$, go to **Step 3**; Otherwise, $\mathbf{T} = \mathbf{Tf}$, $Flag_{uninfor}^i = 0$ and go to **Step 4**.

Step 3: If individual i knows about the location of the temporary target $\mathbf{Tt}^j$ (i.e., $i \in \text{Infor}_j$), $Flag_{uninfor}^i = 0$ and $\mathbf{T} = \mathbf{Tt}^j$; Otherwise, $Flag_{uninfor}^i = 1$.

Step 4: If i sensed obstacle j' (i.e., $\rho(X^i, obs_{j'}) \leq R_{sense}$), $Flag_{obs}^i = j'$. If individual j is a one-hop neighbor of i (i.e., $\|X_{i,2}^{ij}\| \leq R_{comm}$)

and $Flag_{obs}^j = 0$, $Flag_{obs}^j = Flag_{obs}^i$.

Step 5: If $Flag_{uninfor}^i = 0$, the influence weights of neighbors are set to be 1, i.e., $w_j = 1$; Otherwise, ignore the effect from the final target and increase the dependence on neighbors in upper

ranks, i.e., $K_t = 0$ and $\begin{cases} w_j = 1, & Rank_j \leq Rank_i \\ w_j = w', & Rank_j > Rank_i \end{cases}$ where $w' \geq 1$ is

shown by the solid lines in Fig. 1 and $w' = 1$ is equivalent to the flight based on the egalitarian interaction mode.

Step 6: Drive itself based on the pigeon flocking model in Section 3, and then go to **Step 1**.

4. UAV model

In this paper, each UAV is equipped with a first-order march-hold autopilot, a first-order heading-hold autopilot and a second-order altitude-hold autopilot [18,19]. The UAV model is given by the following equation:

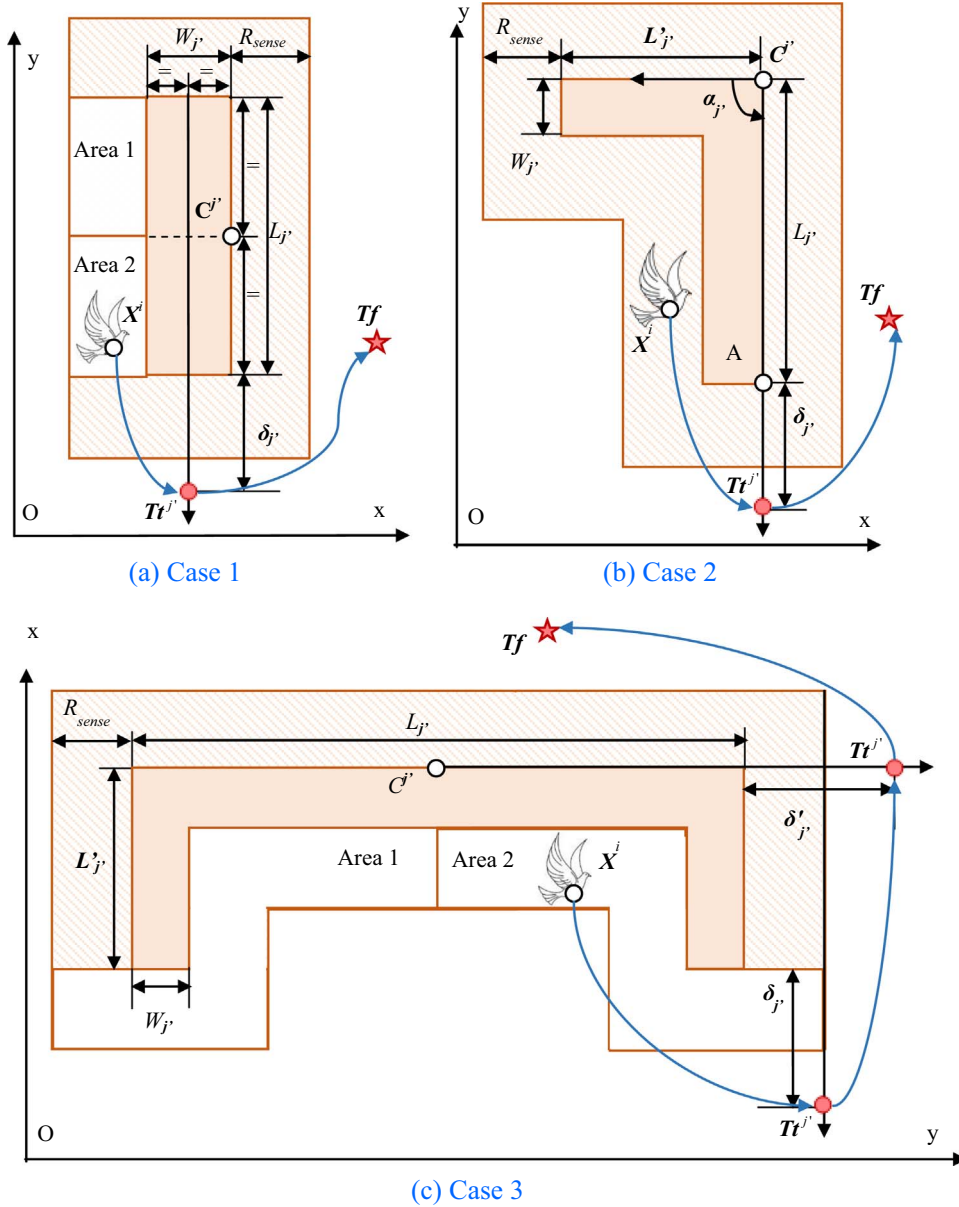


Fig. 2. Locations of temporary targets.

$$\begin{cases} \dot{x}^i = V^i \cos \psi^i \\ \dot{y}^i = V^i \sin \psi^i \\ \dot{h}^i = \lambda^i \\ \dot{V}^i = \frac{1}{\tau_v}(V_c^i - V^i) \\ \dot{\psi}^i = \frac{1}{\tau_\psi}(\psi_c^i - \psi^i) \\ \dot{\lambda}^i = -\frac{1}{\tau_\lambda} \lambda^i + \frac{1}{\tau_h}(h_c^i - h^i) \end{cases} \quad (15)$$

where V^i , ψ^i and λ^i are the horizontal component of airspeed, yaw angle and altitude rate of the UAV respectively, V_c^i , ψ_c^i and h_c^i are the reference signals of the UAV march-hold autopilot, heading-hold autopilot and altitude-hold autopilot respectively, τ_v , τ_ψ and (τ_λ, τ_h) are the time constants of the three autopilots and (x^i, y^i, h^i) is the position of UAV i in the inertial coordinate system.

Considering the applicability, the dynamic constraints are as the following equations:

$$\begin{cases} V_{\min} \leq V^i \leq V_{\max} \\ |\dot{\psi}^i| \leq n_{\max} g / V^i \\ \lambda_{\min} \leq \lambda^i \leq \lambda_{\max} \end{cases} \quad (16)$$

where $V_{\max}$, $V_{\min}$, $n_{\max}$, $\lambda_{\max} > 0$ are the maximum horizontal velocity, minimum horizontal velocity, maximum lateral overload and maximum altitude rate of the UAV respectively, $\lambda_{\min} < 0$ is the minimum altitude rate of the UAV and $g = 10 \text{ m/s}^2$ is the acceleration of gravity.

5. UAV flocking control based on pigeon interaction mode switch

The pigeon flocking and coordinated obstacle-avoiding models established in Sections 2 and 3 are available to be applied to the

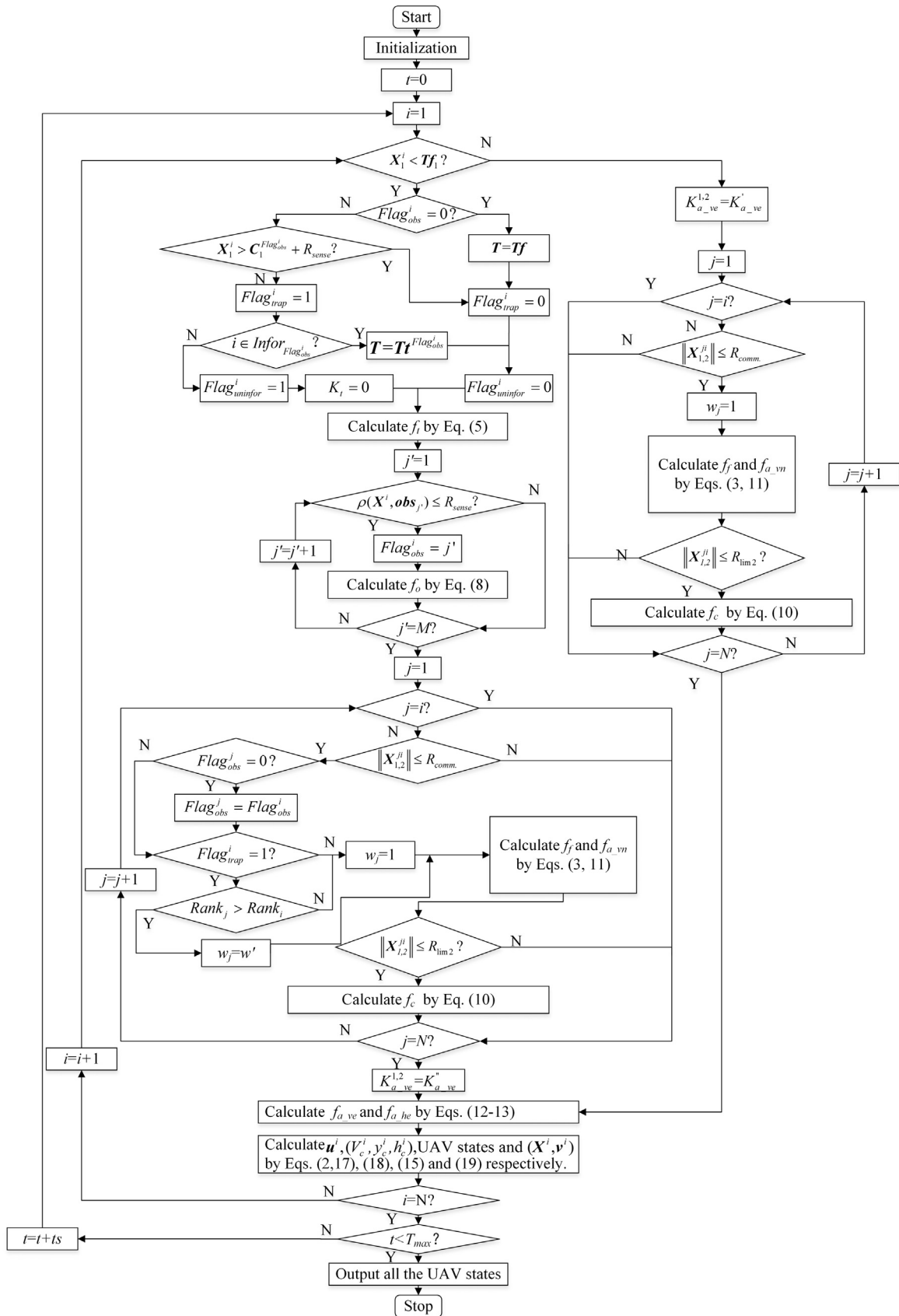


Fig. 3. Process of UAV flocking control based on pigeon interaction mode switch.

Table 1
Parameters of a single UAV.

Variables	Descriptions	Values
τ_v	Time constant of the march-hold autopilot	1 s
τ_ψ	Time constant of the heading-hold autopilot	0.75 s
(τ_a, τ_h)	Time constants of the altitude-hold autopilot	(0.3,1)s
$(v_{\max}, v_{\min})$	Constraints of the horizontal velocity	(5,1)m/s
$n_{\max}$	maximum lateral overload	10 g
$(\lambda_{\max}, \lambda_{\min})$	Constraints of the altitude rate	(5,-5)m/s

Table 2
Parameters of obstacles.

j'	Case	$C^j(m)$	$L_j(m)$	$L'_j(m)$	$W_j(m)$	$\alpha_j(rad)$	Infor $_j$
1	1	(60, 0)	60	/	10	/	{1,2,3,4,5}
2	2	(120, 80)	40	30	10	$3\pi/2$	{1,2,3}
3	2	(120, -80)	40	30	10	$\pi/2$	{1,2,3}
4	3	(200, 0)	120	30	10	/	{1}

distributed flocking control under obstacle environments for a UAV swarm by treating each UAV as a pigeon in the flock. As shown in Fig. 3, the distributed flocking control algorithm for a UAV swarm is implemented as the following steps:

Step 1: Initialize the states of N UAVs, including the position, horizontal component of airspeed, yaw angle and altitude rate. Initialize identifiers $Flag_{obs}^i$, $Flag_{trap}^i$ and $Flag_{uninfor}^i$ as 0.

Step 2: $t = 0$ where t is current simulation time.

Step 3: $i = 1$.

Step 4: If individual i have passed the final target in the x -axis direction (i.e. $X_1^i \geq T_f$), $K_{a_ve}^{1,2} = K_{a_ve}'$ and go to **Step 12** where K_{a_ve}' is the desired velocity alignment gain in the x -axis and y -axis directions when i have passed the final target; Otherwise, go to **Step 5**.

Step 5: If $Flag_{obs}^i = 0$, $T = T_f$. If $Flag_{obs}^i \neq 0$ and $X_1^i \leq C_1^{Flag_{obs}^i} + R_{sense}$, $Flag_{trap}^i = 1$; Otherwise, $Flag_{trap}^i = 0$ and $Flag_{uninfor}^i = 0$. If $Flag_{trap}^i = 1$ and $i \in Infor_{Flag_{obs}^i}$, $T = T_t^{Flag_{obs}^i}$ and $Flag_{uninfor}^i = 0$; Otherwise, $Flag_{uninfor}^i = 1$ and $K_t = 0$. Calculate f_t by Eq. (5).

Step 6: $j' = 1$.

Step 7: If $\rho(X^i, obs_{j'}) \leq R_{sense}$, $Flag_{obs}^i = j'$ and calculate f_o by Eq. (8).

Step 8: If $j' = M$, go to **Step 9**; otherwise $j' = j' + 1$, then go to **Step 7**.

Step 9: $j = 1$.

Step 10: If $j \neq i$, $\|X_{1,2}^{ji}\| \leq R_{comm}$ and $Flag_{obs}^j = 0$, $Flag_{obs}^j = Flag_{obs}^i$. If

$j \neq i$, $\|X_{1,2}^{ji}\| \leq R_{comm}$ and $Flag_{trap}^i = 1$, $\begin{cases} w_j = 1, & Rank_j \leq Rank_i \\ w_j = w', & Rank_j > Rank_i \end{cases}$. If

$j \neq i$, $\|X_{1,2}^{ji}\| \leq R_{comm}$ and $Flag_{trap}^i = 0$, $w_j = 1$. Calculate f_f and f_{a_vn}

by Eqs. (3) and (11) respectively. If $j \neq i$ and $\|X_{1,2}^{ji}\| \leq R_{lim2}$,

calculate f_c by Eq. (10).

Step 11: If $j = N$, $K_{a_ve}^{1,2} = K_{a_ve}''$ go to **Step 15** where K_{a_ve}'' is the desired velocity alignment gain in the x -axis and y -axis directions before passing the final target; otherwise $j = j + 1$, then go to **Step 10**.

Step 12: $j = 1$.

Step 13: If $j \neq i$ and $\|X_{1,2}^{ji}\| \leq R_{comm}$, $w_j = 1$ and calculate f_f and f_{a_vn} by Eqs. (3) and (11) respectively. If $j \neq i$ and $\|X_{1,2}^{ji}\| \leq R_{lim2}$, calculate f_c by Eq. (10).

Step 14: If $j = N$, go to **Step 15**; otherwise $j = j + 1$, then go to **Step 13**.

Step 15: Calculate f_{a_ve} and f_{a_he} by Eqs. (12) and (13) respectively.

Step 16: Calculate u^i by Eq. (2) and the flowing equation:

$$u^i = \begin{cases} u^i, & u^i \geq u_{lim} \\ 0, & u^i < u_{lim} \end{cases} \quad (17)$$

where u_{lim} is dead zone threshold of control inputs. Calculate (V_c^i, y_c^i, h_c^i) by the following equation:

$$\begin{cases} V_c^i = \tau_v(u_1^i \cos \psi^i + u_2^i \sin \psi^i) + V^i \\ \psi_c^i = \frac{\tau_\psi}{V^i}(u_2^i \cos \psi^i - u_1^i \sin \psi^i) + \psi^i \\ h_c^i = h^i + \frac{\tau_h \lambda}{\tau_\lambda} + \tau_h u_3^i \end{cases} \quad (18)$$

Calculate the UAV states at next time by Eq. (15). Calculate the position vector and velocity vector of pigeon i by the flowing equation:

$$\begin{cases} X^i = (x^i, y^i, h^i) \\ v^i = (V^i \cos \psi^i, V^i \sin \psi^i, h^i) \end{cases} \quad (19)$$

Step 17: If $i = N$, go to **Step 18**; otherwise $i = i + 1$, then go to **Step 4**.

Step 18: If $t < T_{\max}$, $t = t + t_s$ and then go to **Step 3**, where $T_{\max}$ is the maximum simulation time and t_s is the sampling time. Otherwise output the states of N UAVs.

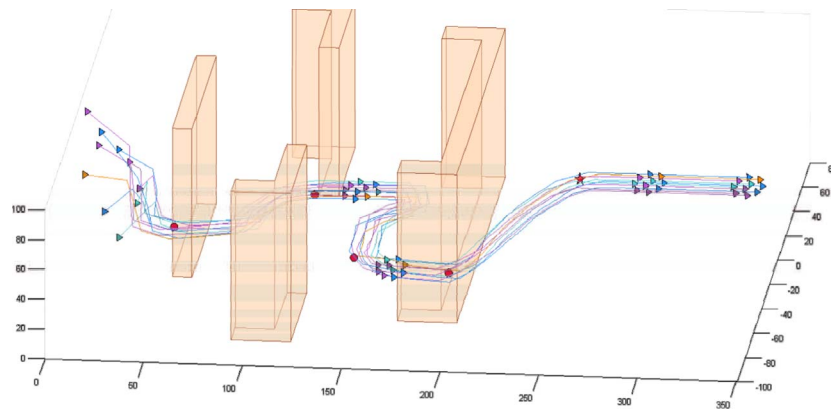
6. Comparative simulation results

To validate the effectiveness of the distributed flocking control algorithm for a UAV swarm based on the pigeon interaction mode switch behavior, 10 UAVs are disposed to fly through a complex environment with obstacles.

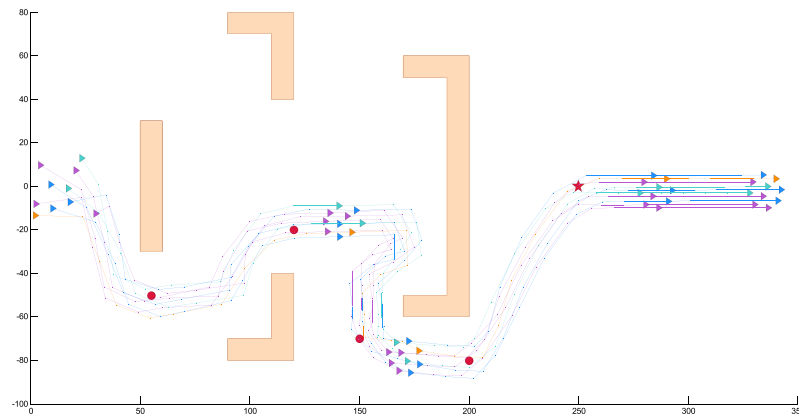
The parameters of each UAV are shown in Table 1. In the UAV swarm's forward direction to the final target $T_f = (250m, 0m, 50m)$, four 100-meter-high obstacles exist with the parameters shown in Table 2. The potential field influence

Table 3
Initial parameters of the UAV swarm.

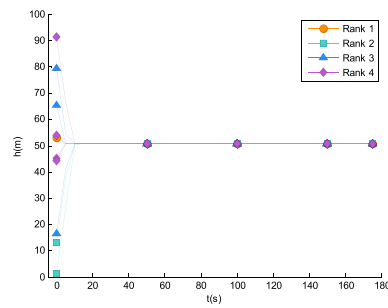
i	1	2	3	4	5	6	7	8	9	10
$Rank_i$	1	2	2	3	3	4	4	4	4	4
$x^i(m)$	2.28	23.38	17.06	10.11	9.34	18.06	20.68	2.51	4.57	29.88
$y^i(m)$	-13.38	13.02	-0.92	-10.13	0.86	-7.11	7.44	-8.13	9.77	-12.65
$h^i(m)$	53.08	12.99	1.19	79.43	16.56	65.41	45.05	91.33	53.83	44.27



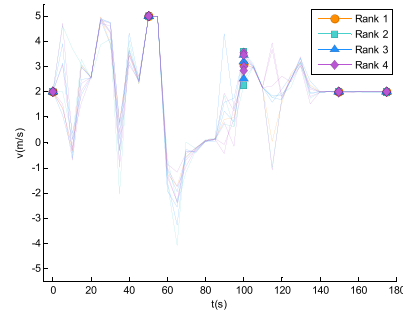
(a) Collective motion trajectories



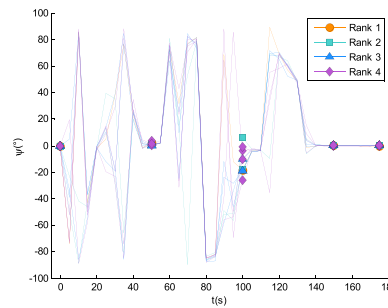
(b) Collective motion trajectories in top-down view



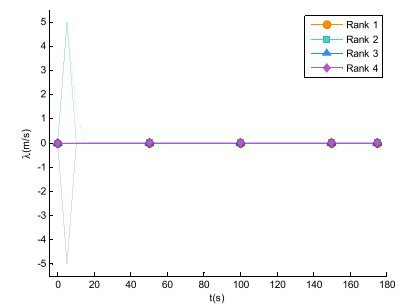
(c) Altitudes of UAVs



(d) Horizontal velocities of UAVs



(e) Yaw angles of UAVs



(f) Altitude rates of UAVs

Fig. 4. UAV distributed flocking based on pigeon interaction mode switch.

limited distance R_{sense} of each obstacle is 10m. As shown in Tables 2 and 3, the UAVs in the swarm are divided into four ranks: UAV 1 in the hierarchical rank 1 is informed about how to set temporary targets for all cases of obstacles, UAVs 2 and 3 in the

rank 2 are informed about temporary targets for cases 1 and 2, UAVs 4 and 5 in the rank3 are informed about temporary targets for case 1, and UAVs 6–10 are uninformed about how to set temporary targets. When uninformed about the temporary target for

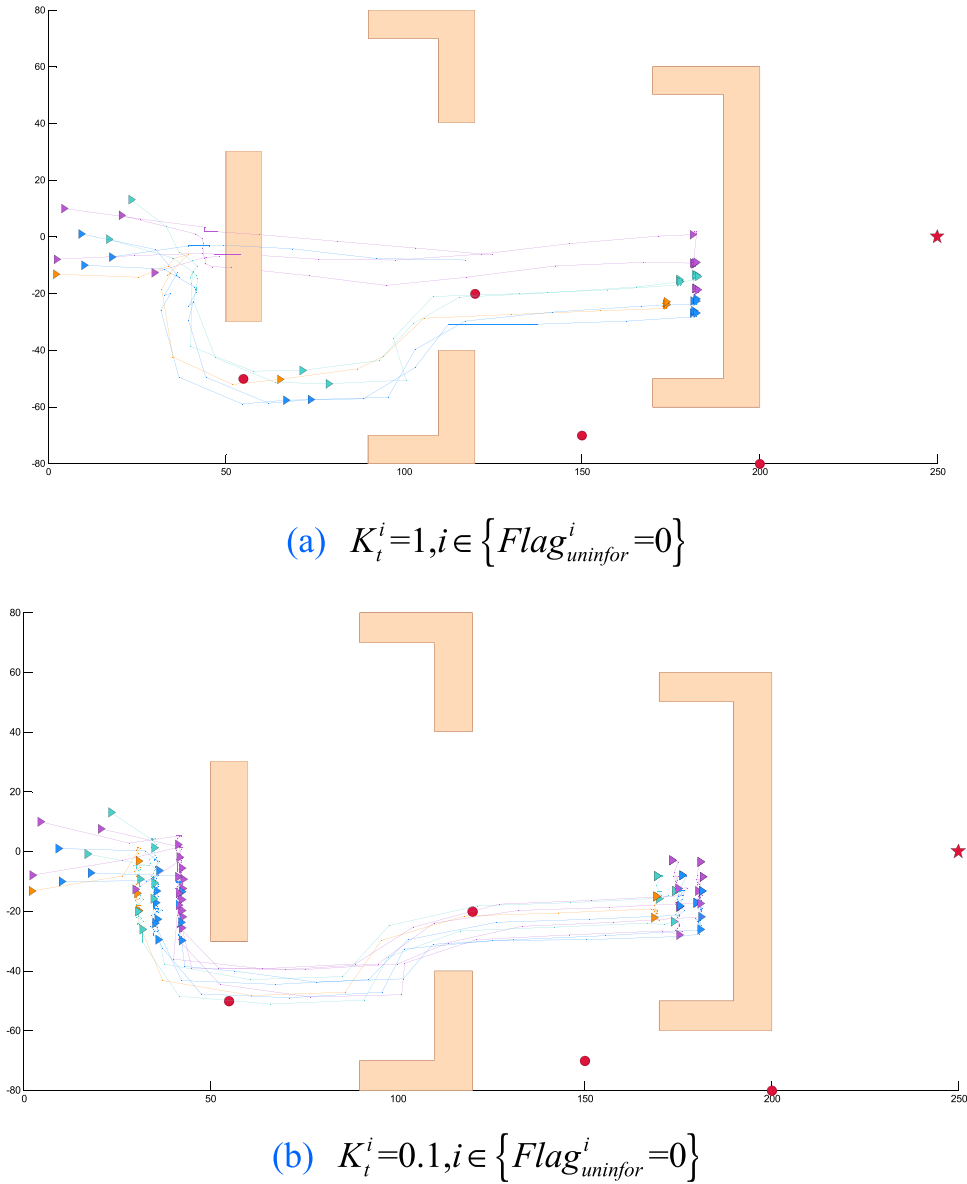


Fig. 5. UAV distributed flocking based on pigeon egalitarian interaction mode without $K_t^i = 0, i \in \{Flag_{uninfor}^i = 0\}$.

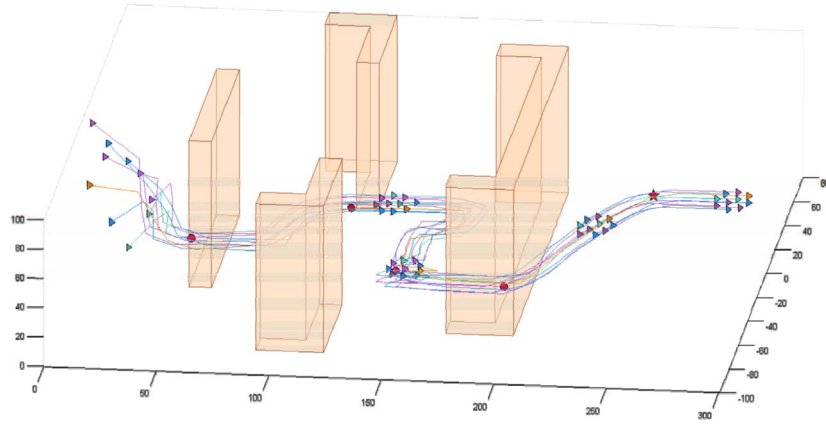
current obstacle, UAVs 2 and 3, UAVs 4 and 5 and UAVs 6–10 will put particular emphasis on UAV 1, UAVs 1–3 and UAVs 1–5 respectively. As shown in Table 3, the initial positions (x, y, h) of the UAV swarm are initialized randomly. The initial horizontal component of airspeed V , yaw angle ψ and altitude rate λ of the UAV swarm are $2m/s$, $0rad$ and $0m/s$ respectively. The horizontal communication range R_{comm} of each UAV is $25m$.

The simulation is conducted according to the specific steps of the proposed UAV distributed flocking control algorithm in Section 5, where the maximum simulation time $T_{max} = 175s$, the sampling time $ts = 0.05s$, and the control input dead zone threshold $u_{lim} = 0.1N$. The parameters of the pigeon flocking model are as follows: The formation control gain K_f , target attraction gain K_t , obstacle repulsion gain K_o , neighbor repulsion gain K_c , neighbor velocity alignment gain $K_{a_{vn}}$, desired velocity alignment gains ($K'_{a_{ve}}, K''_{a_{ve}}$) and $K^3_{a_{ve}}$, and target altitude alignment gain $K_{a_{he}}$ are 10, 0.01, 100,000, 100,000, 1, (2, 0.3), 1 and 70 respectively, the expected horizontal distance R_{desire} is $10m$, the target's potential field influence limited distance R_{lim1} is $1m$, the individual repulsion potential field influence limited distance R_{lim2} is $2m$ and the desired velocity $\mathbf{ve}$ is $(2, 0, 0)m/s$. The parameters of the pigeon

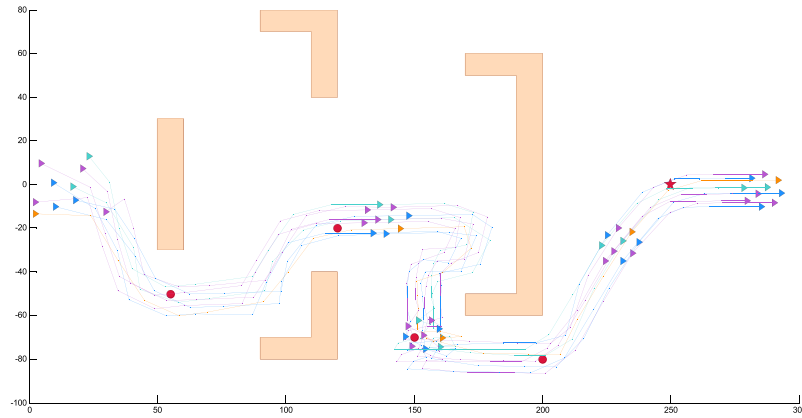
coordinated obstacle-avoiding model are as follows: The influence weight w' of individuals in the upper rank to individuals in the lower rank is 2, and the temporary target setting parameters δ_j are all $20m$.

Fig. 4 depicts the detailed results of the distributed flocking flight for the UAV swarm under the obstacle environment, in which (a)–(f) describe three-dimensional collective motion trajectories, collective motion trajectories in top-down view, altitudes, horizontal velocities, yaw angles and altitude rates of the UAV swarm respectively. Fig. 4(a) and (b) illustrates that the UAVs in the swarm are qualified to flight safely through the obstacle environment. In the whole flight process, the UAVs are always in a swarm without collisions, and fly in a stable formation when passed the final target. As shown in Fig. 4(c) and (f), the altitudes and altitude rates of UAVs respectively converge to the target altitude and $0m/s$ at about 10 s. As shown in Fig. 4(d) and (e), the horizontal velocities and yaw angles of UAVs respectively converge to $2m/s$ and $0rad$ at about 140 s after fluctuations within the allowable range.

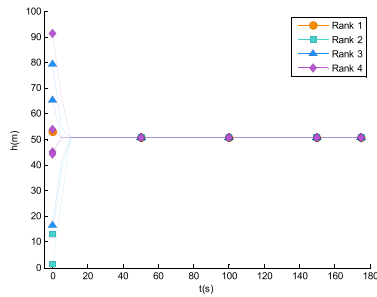
In order to further illustrate the superiority of the proposed distributed flocking control algorithm, the UAV swarm flight



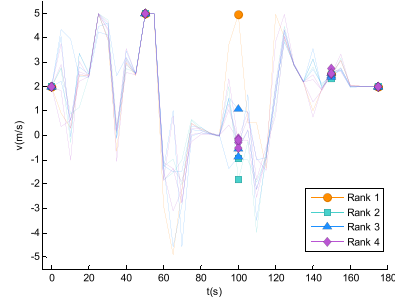
(a) Collective motion trajectories



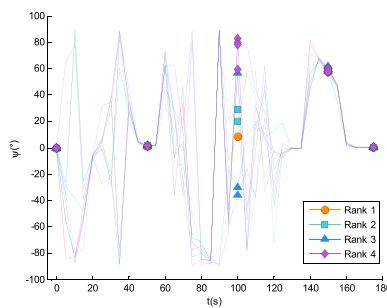
(b) Collective motion trajectories in top-down view



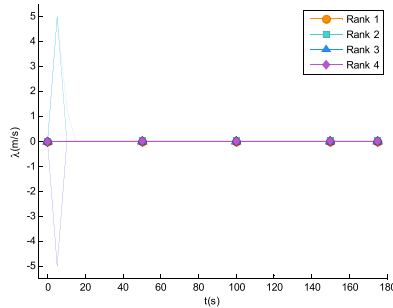
(c) Altitudes of UAVs



(d) Horizontal velocities of UAVs



(e) Yaw angles of UAVs



(f) Altitude rates of UAVs

Fig. 6. UAV distributed flocking based on pigeon egalitarian interaction mode with $K_t^i = 0$, $i \in \{Flag_{unifor}^i = 0\}$.

simulation based on the pigeon egalitarian interaction behavior was conducted with the same initial condition and parameter configuration. If individuals with $Flag_{unifor}^i = 1$ still pay attention to

the effect from the final target, they will be disengaged from the swarm and not avoid obstacles due to the great attraction from the final target (As shown in Fig. 5(a)). If target attraction gain K_t

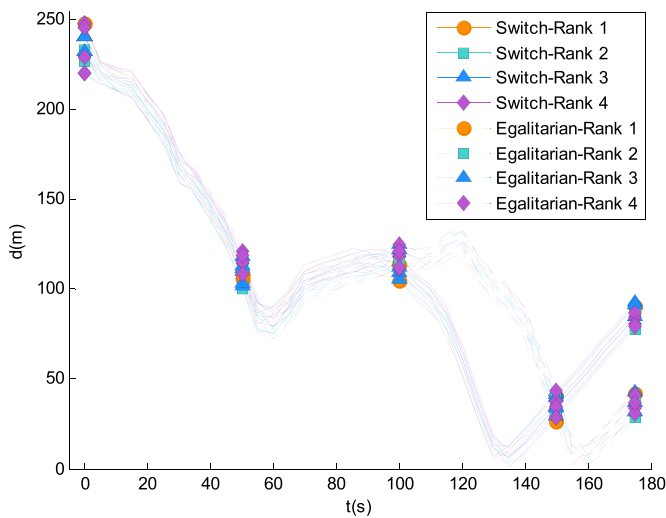


Fig. 7. Distances between individuals and the final target (Switch vs Egalitarian).

decreased to 0.1, the UAV swarm will still fall into the local minimum in front of obstacle 4 due to the insistence on the final target from too many uninformed individuals (As shown in Fig. 5(b)). If uninformed individuals ignore the effect from the final target and don't increase the dependence on neighbors in upper ranks (i.e., $w' = 1$), the UAV swarm can fly safely through the obstacle environment and form a stable formation after passing the final target with some redundant trajectories as shown in Fig. 6(a) and (b). As shown in Fig. 6(d) and (e), the horizontal velocities and yaw angles of UAVs respectively converge to $2m/s$ and $0rad$ at about 165 s after fluctuations within the allowable range and fall behind the proposed distributed flocking algorithm by almost 25 s. Fig. 7 describes the distances between individuals and the final target. The whole UAV swarm (shown by the solid lines) based on the pigeon interaction mode switch passed the final target at about 135 s and was ahead of the UAV swarm (shown by the dashed lines) based on the pigeon egalitarian interaction mode by 25 s.

To sum up, the UAV swarm with few informed individuals is capable to coordinate itself to fly safely through the obstacle environment and finally form a stable formation by means of the proposed distributed flocking control algorithm based on the interaction mode switch behavior of pigeon flocks. Compared to the traditional flocking control algorithm based on the egalitarian interaction mode, the proposed distributed flocking control algorithm has certain superiority in the trajectory smoothness, reaching time to the final target and convergence time of UAV states.

7. Conclusions

The usage of UAV swarms enhances the ability to control battle-field situations, but it also requires significant investments in distributed control technologies due to the booming computational overheads and communication bandwidths. Since the similarity between pigeon flocks and UAV swarms in mission requirements, a distributed flocking control algorithm based on the pigeon interaction mode switch behavior is designed to coordinate a heterogeneous UAV swarm to fly through obstacle environments. The key of the distributed flocking control algorithm design is how to judge currently adopted interaction mode. This paper is a primary exploration of establishing mapping relations between natural and man-made collective systems.

Actually, current observations of a natural collective system are only snapshots that represent the outcome of a long history of previous states [20]. Feedback based on individual learning is a crucial mechanism in understanding natural collective systems. In the future, we will focus on describing the collective behavior in natural systems in a time-depth perspective by considering the memory and learning of individuals. We will also develop more theoretical research on collective systems [21–23] to apply the natural mechanism to UAV swarm control to promote the emergent intelligence of the man-made system.

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